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ORIGINAL RESEARCH

Viral Pathogen-Specific Severe Acute Respiratory Infections in Hospitalised Adults in Europe: Epidemiological Findings from the id.DRIVE Study 2024–2025

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ABSTRACT

Introduction: Lower respiratory tract infections represent a major public health burden, but adult data on respiratory viruses except influenza and COVID-19 are limited. Most European surveillance systems report aggregate data lacking patient-level detail.

Methods: We utilised the multicountry id.DRIVE surveillance study data to describe the epidemiology, clinical characteristics, and outcomes of adults hospitalised with pathogen-specific severe acute respiratory infections (SARI) during 2024–2025. Adults (≥ 18 years)

Prior presentation: Partial findings from this study were presented at the 9th ReSViNET conference (RSVVW'26), February 17–20, 2026, Rome, Italy; and at ESCMID Global 2026, April 17–21, 2026, Munich, Germany.

Collaborators: The members of the id.DRIVE COLLABORATIVE AUTHORING GROUP are listed in the Acknowledgements section, with their detailed affiliations provided in the Electronic Supplementary Materials file.

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hospitalised with SARI across 17 hospitals in Spain, Italy, and Germany between 1 September 2024 and 31 August 2025 were included. All participants were tested using reverse transcription polymerase chain reaction for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), respiratory syncytial virus (RSV), and influenza A/B (testing additional respiratory pathogens varied by hospital). Descriptive analyses were stratified by pathogen.

Results: Of 8730 included participants (median age, 77 years), 14.3% (n = 1249) tested positive for influenza A, 6.9% (n = 606) SARS-CoV-2, 6.4% (n = 554) RSV, and 0.8% (n = 74) for influenza B. Positivity peaked for influenza and RSV in December–January, and for SARS-CoV-2 in July. Comorbidity burden was substantial: 51.9% of all SARI participants had ≥ 3 chronic conditions, reaching 62.3% in those with RSV. RSV-positive SARI patients were older (median age, 80 years); oxygen therapy was required in 72.0%, non-invasive ventilation in 12.3%, and in-hospital mortality was 9.4%. Over 75.0% of SARI cases positive for a respiratory virus received antibiotics during hospitalisation.

Conclusion: Viral pathogens remain a major cause of adult SARI hospitalisations in Europe. Our findings highlight the continued burden and seasonal diversity of viral respiratory pathogens, supporting the value of integrated, pathogen-specific SARI surveillance guiding prevention strategies.

Keywords: SARI; Surveillance; Epidemiology; SARS-CoV-2; Influenza; RSV; HMPV; Hospitalisation

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Key Summary Points

Why carry out this study?

Data on the pathogen-specific epidemiology, clinical characteristics, and outcomes of hospitalised adults with severe acute respiratory infections (SARI) in Europe remain limited, particularly for respiratory viruses other than SARS-CoV-2 and influenza.

We aimed to characterise laboratory-confirmed viral infections among adults hospitalised with SARI, using data from a multicountry hospital-based SARI surveillance network (id.DRIVE).

What was learned from the study?

During the 2024–2025 season, influenza A was the most frequently detected respiratory virus among adults hospitalised with SARI across participating hospitals.

Most SARI patients were aged ≥ 75 years with multiple comorbidities, and respiratory syncytial virus cases had the highest proportion of multimorbidity as well as higher in-hospital mortality than other respiratory virus cases.

Pathogen-specific SARI surveillance linking patient-level laboratory and clinical data can supplement existing surveillance by improving characterisation of pathogen distribution and clinical outcomes among hospitalised adults, including for respiratory viruses not routinely monitored.

INTRODUCTION

Lower respiratory tract infections (LRTIs) are a leading cause of global morbidity and mortality [1]. A wide range of viruses contribute to the burden of LRTIs, including influenza, severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), respiratory syncytial virus (RSV), human metapneumovirus (HMPV), and parainfluenza viruses (PIVs). The burden of influenza [2], and COVID-19 [3] are well documented. However, the burden of RSV, although increasingly recognised [4], remains largely underestimated due to limited routine testing [5, 6], variable case definitions [6], testing periods that do not always align with the RSV season, and suboptimal sensitivity of commonly used assays [7, 8]. For other viral respiratory pathogens, such as HMPV, PIVs, and adenoviruses, data on severe acute respiratory infection (SARI)-related epidemiology and clinical characteristics in adults remain more limited.

Following the COVID-19 emergency, other viral causes of LRTIs have resurged, with seasonal RSV and influenza incidence now matching or exceeding pre-pandemic levels [9, 10], reinforcing the need for strengthened SARI surveillance. Sentinel surveillance networks can monitor SARI by collecting patient-level data to track epidemiological trends, identify vulnerable populations, and assess vaccine effectiveness, activities that remain important as broader global investment in public health preparedness declines post-pandemic [11]. Several European SARI surveillance systems were established following the COVID-19 pandemic [12]; however, most do not routinely link laboratory results with patient-level clinical and outcome data [12], limiting analyses of pathogen-specific epidemiology and disease burden. In addition, current surveillance systems do not always incorporate multiplex polymerase chain reaction (PCR) results, limiting information on pathogens beyond influenza and SARS-CoV-2.

To address these gaps, we utilised data from the id.DRIVE network, for the 2024–2025 season, to describe the epidemiology, clinical characteristics, and outcomes of adults hospitalised with SARI who were enrolled into the id.DRIVE surveillance study across three European countries. Analyses were conducted both overall and stratified by laboratory-confirmed viral pathogens. The primary pathogens were SARS-CoV-2, influenza A and B, and RSV. In addition, adenoviruses, PIVs (types 1–4), HMPV, human rhinovirus (HRV)/human enterovirus (HEV), and seasonal human coronaviruses (229E, NL63, OC43,

and HKU1) were also included from a subset of 14 of the included 17 hospitals that had collected data on these pathogens. Our analysis aims to provide a descriptive overview of pathogen-specific SARI burden within the surveillance network rather than a formal validation of the platform against other data sources, although it may complement existing European surveillance systems.

METHODS

Study Design and Setting

The id.DRIVE surveillance study is an ongoing multicountry surveillance study monitoring viral respiratory infections among adults hospitalised with SARI via a sentinel network of participating hospitals in Europe [13]. For the present analysis, we included nine Study Contributors (Supplementary Table S1), representing participating sites that are either hospitals or hospital networks distributed across Spain, Italy, and Germany. We included participants enrolled at these sites covering 17 tertiary-level hospitals between 1 September 2024 and 31 August 2025. Four participating id.DRIVE sites and one hospital from one participating site were excluded from the analysis due to limited recruitment durations during the study period.

Study Participants

The study population included adults (≥ 18 years) admitted to participating hospitals during the study period. Eligible participants were hospitalised for at least one overnight stay with SARI. SARI was defined based on clinical suspicion of a respiratory infection with at least one of the following symptoms: cough, fever, shortness of breath, sudden onset of anosmia, ageusia, or dysgeusia. Symptom onset had to occur within 14 days prior to hospital admission. This SARI definition was modified from the latest European Centre for Disease Prevention and Control (ECDC) case definition by specifying "suspicion of respiratory infection" [14]. Eligible SARI patients were identified prospectively (e.g. during consultation in the emergency department or at admission to an inpatient ward). At 3 of the 17 included hospitals, eligible SARI patients were identified retrospectively by querying electronic hospital databases. Recruitment settings varied across sites; 5 participating sites covering 13 (76.0%) hospitals were unable to recruit patients while they were in the intensive care unit (ICU) (Supplementary Table S1). Patients admitted directly to the ICU could be enrolled following transfer to an inpatient ward.

Laboratory Testing

Respiratory specimens were collected from all eligible participants either during routine clinical practice or specifically for the study. While nasopharyngeal or oropharyngeal swabs were recommended in the study protocol, other types of respiratory specimens were also accepted. Molecular testing for SARS-CoV-2, influenza A/B, and RSV was conducted for all participants using reverse transcription PCR, or an equivalent nucleic acid amplification test with equivalent sensitivity, on respiratory specimens collected within 14 days prior to and up to 72 h after admission. Fourteen of the 17 included hospitals also conducted PCR testing for additional respiratory pathogens of interest, including RSV A/B, and other pathogens such as adenovirus, PIVs (types 1–4), HMPV, HRV/HEV, human bocavirus, and other human coronaviruses (229E/NL63/OC43/HKU1), with a varied number of pathogens tested across study contributors (Supplementary Table S2).

Data Collection

Data were collected on participants enrolled into id.DRIVE prospectively from 1 September 2024 (earliest sites) to 31 August 2025; however, the exact recruitment start dates of sites varied throughout the study period (Supplementary Table S1). Three participating sites collected data retrospectively on eligible SARI patients during the study period (Supplementary Table S1).

Data sources included hospital medical records, vaccine registries, general practitioner medical records, and prospectively collected study data. Demographic, clinical, and outcome data were collected from hospital medical records and entered into an electronic Case Report Form (eCRF, Castor®), with only pseudonymised data being transferred from the Study Contributors to the id.DRIVE research server, a dedicated and secured central server hosted by P95 (study sponsor).

The following variables were included: age at admission; sex; number and type of chronic conditions; long-term care facility residence; smoking history; ever pneumococcal vaccine ≥ 14 days prior to symptom onset; at least one dose of influenza vaccine and/or SARS-CoV-2 vaccine within the 12 months preceding hospitalisation; symptom onset date; chronic conditions ongoing at the time of hospitalisation, except for cancer that was either in treatment, in active follow-up, or within 5 years since post-curative treatment (Supplementary Table S3); and treatments received for current SARI episode (pre- and post-hospital admission).

Additionally, clinical outcome data were collected from all participants and included: the highest level of respiratory support received; length of hospital stay; ICU admission; discharge destination; and in-hospital mortality.

COVID-19 and influenza vaccination status was defined as receipt of at least one vaccine dose ≥ 14 days before symptom onset date within the relevant country-specific vaccination season. For pneumococcal vaccination status, participants were considered vaccinated if they had ever received the respective vaccine at least 14 days before symptom onset, regardless of season. RSV vaccination status is not reported, as RSV vaccination was available only in Germany during the study period, and Germany contributed only a small proportion of reported SARI cases.

Statistical Methods

Descriptive analyses were stratified by pathogen status and co-infection prevalence. Results are presented as counts and percentages for categorical variables, and as medians (25th–75th percentiles) for continuous variables. Weekly test positivity rates were calculated as the proportion of SARI participants with laboratory-confirmed SARS-CoV-2, influenza A/B, or RSV infection among all SARI participants tested for these four pathogens, based on the date of symptom onset. To assess potential selection bias in ICU recruitment, descriptive analyses of clinical outcomes were further stratified according to whether participants were enrolled from sites with direct ICU enrolment of participants.

This study was performed in accordance with the 1964 Declaration of Helsinki and its later amendments. All data were pseudonymised prior to analysis and processed in accordance with the European Union General Data Protection Regulation [GDPR; Regulation (EU) 2016/679]. The id.DRIVE surveillance protocol (EUPAS1000000012) was approved by independent ethics committees (IECs) at all participating hospitals or hospital networks: Comitato Etico Regionale della Liguria (Genoa); Comité Ético de Investigación con Medicamentos del Departamento de Salud Arnau de Vilanova (Valencia); Comité Ético de Investigación con Medicamentos del Hospital Universitari Vall d'Hebron (Barcelona); Comité de Ética de la Investigación con-Medicamentos del Hospital Universitario La Paz; Ethikkommission des Fachbereichs Medizin der Goethe-Universität(Frankfurt); Ethikkommission Ulm. Individuals (or their legally acceptable representative) provided informed consent when applicable. Waiver of informed consent was authorised by local IECs for retrospective data collection for Hospital Germans Trias i Pujol, Hospital Clinic Barcelona, and Vall d'Hebron, as the tests carried out and information collected were part of routine clinical practice and epidemiological surveillance.

RESULTS

Study Population

Among 8959 SARI participants enrolled in id.DRIVE from 17 hospitals during the study period, 229 were excluded for lack of an

Table 1 Characteristics of SARI patients, by SARS-CoV-2, influenza, and RSV test result. id.DRIVE, 1 September 2024–31 August 2025

	All SARI patients *n* (%)	SARS-CoV-2 cases *n* (%)	Influenza cases *n* (%)			RSV cases[c] *n* (%)
			Total[b]	Influenza A	Influenza B	
Total[a] (*N*)	8730	606	1249	1177	74	554
Sex						
Male	4482 (51.3%)	305 (50.3%)	604 (48.4%)	569 (48.3%)	36 (48.6%)	244 (44.0%)
Female	4248 (48.7%)	301 (49.7%)	645 (51.6%)	608 (51.7%)	38 (51.4%)	310 (56.0%)
Age, years, median (lower, upper quartile)	77 (65, 86)	80 (69, 87)	76 (63, 85)	76 (64, 85)	58.5 (37.25, 74)	80 (71, 88)
Age group						
18–49 years	730 (8.4%)	45 (7.4%)	137 (11.0%)	105 (8.9%)	33 (44.6%)	18 (3.2%)
50–59 years	729 (8.4%)	34 (5.6%)	114 (9.1%)	108 (9.2%)	6 (8.1%)	33 (6.0%)
60–74 years	2386 (27.3%)	143 (23.6%)	334 (26.7%)	318 (27.0%)	17 (23.0%)	131 (23.6%)
≥ 75 years	4885 (56.0%)	384 (63.4%)	664 (53.2%)	646 (54.9%)	18 (24.3%)	372 (67.1%)
Country						
Germany	179 (2.1%)	18 (3.0%)	24 (1.9%)	22 (1.9%)	2 (2.7%)	6 (1.1%)
Italy	768 (8.8%)	83 (13.7%)	110 (8.8%)	100 (8.5%)	10 (13.5%)	36 (6.5%)
Spain	7783 (89.2%)	505 (83.3%)	1115 (89.3%)	1055 (89.6%)	62 (83.8%)	512 (92.4%)
Symptom onset to specimen collection,[d] days						
Number of days, median (lower, upper quartile)	3 (2, 7)	3 (1, 5)	3 (2, 6)	3 (2, 6)	4.5 (2, 7)	4 (2, 6)
Number of chronic conditions[e]						
0	640 (7.3%)	29 (4.8%)	122 (9.8%)	104 (8.8%)	19 (25.7%)	21 (3.8%)
1	1433 (16.4%)	87 (14.4%)	209 (16.7%)	195 (16.6%)	14 (18.9%)	63 (11.4%)
2	2022 (23.2%)	152 (25.1%)	312 (25.0%)	298 (25.3%)	14 (18.9%)	116 (20.9%)
≥ 3	4533 (51.9%)	333 (55.0%)	600 (48.0%)	575 (48.9%)	26 (35.1%)	345 (62.3%)
Missing	102 (1.2%)	5 (0.8%)	6 (0.5%)	5 (0.4%)	1 (1.4%)	9 (1.6%)
Number of chronic conditions, median (lower, upper quartile)	3 (2, 4)	3 (2, 4)	2 (1, 4)	2 (1, 4)	2 (0, 3)	3 (2, 4)

Table 1 continued

	All SARI patients *n* (%)	SARS-CoV-2 cases *n* (%)	Influenza cases *n* (%)			RSV cases[c] *n* (%)
			Total[b]	Influenza A	Influenza B	
Chronic condition[e]						
Asthma	760 (8.7%)	38 (6.3%)	129 (10.3%)	120 (10.2%)	10 (13.5%)	71 (12.8%)
Lung disease	3105 (35.6%)	180 (29.7%)	368 (29.5%)	353 (30.0%)	15 (20.3%)	198 (35.7%)
COPD	2084 (67.1%)	109 (60.6%)	233 (63.3%)	227 (64.3%)	6 (40.0%)	132 (66.7%)
Cardiovascular disease	3759 (43.1%)	303 (50.0%)	499 (40.0%)	480 (40.8%)	20 (27.0%)	305 (55.1%)
Congestive heart failure	1071 (28.5%)	98 (32.3%)	136 (27.3%)	131 (27.3%)	5 (25.0%)	76 (24.9%)
Hypertension	5189 (59.4%)	385 (63.5%)	715 (57.2%)	691 (58.7%)	25 (33.8%)	383 (69.1%)
Chronic renal disease	1665 (19.1%)	151 (24.9%)	218 (17.5%)	207 (17.6%)	11 (14.9%)	130 (23.5%)
End-stage kidney disease	416 (25.0%)	39 (25.8%)	57 (26.1%)	54 (26.1%)	3 (27.3%)	38 (29.2%)
Chronic liver disease	571 (6.5%)	30 (5.0%)	82 (6.6%)	78 (6.6%)	4 (5.4%)	29 (5.2%)
End-stage liver disease	24 (4.2%)	0 (0.0%)	4 (4.9%)	4 (5.1%)	0 (0.0%)	0 (0.0%)
Obesity	1920 (22.0%)	121 (20.0%)	251 (20.1%)	239 (20.3%)	13 (17.6%)	145 (26.2%)
Type 1 diabetes	53 (0.6%)	3 (0.5%)	5 (0.4%)	4 (0.3%)	1 (1.4%)	7 (1.3%)
Type 2 diabetes	2515 (28.8%)	176 (29.0%)	339 (27.1%)	329 (28.0%)	11 (14.9%)	169 (30.5%)
Neurological disorders	1861 (21.3%)	154 (25.4%)	258 (20.7%)	249 (21.2%)	9 (12.2%)	124 (22.4%)
Cancer	1693 (19.4%)	147 (24.3%)	222 (17.8%)	211 (17.9%)	11 (14.9%)	102 (18.4%)
Solid tumour	1288 (76.1%)	101 (68.7%)	175 (78.8%)	171 (81.0%)	4 (36.4%)	64 (62.7%)
Haematological cancer	417 (24.6%)	46 (31.3%)	48 (21.6%)	41 (19.4%)	7 (63.6%)	38 (37.3%)
Immunodeficiency[f]	607 (7.0%)	56 (9.2%)	74 (5.9%)	66 (5.6%)	9 (12.2%)	33 (6.0%)
Cancer and/or immunodeficiency[g]	1978 (22.7%)	168 (27.7%)	265 (21.2%)	250 (21.2%)	16 (21.6%)	115 (20.8%)
Vaccination status,[h] yes						
COVID-19	4874 (55.8%)	334 (55.1%)	608 (48.7%)	588 (50.0%)	21 (28.4%)	371 (67.0%)
Influenza	4361 (50.0%)	294 (48.5%)	540 (43.2%)	525 (44.6%)	16 (21.6%)	333 (60.1%)
Pneumococcal disease	3943 (45.2%)	291 (48.0%)	539 (43.2%)	519 (44.1%)	21 (28.4%)	302 (54.5%)

Table 1 continued

	All SARI patients *n* (%)	SARS-CoV-2 cases *n* (%)	Influenza cases *n* (%)			RSV cases[c] *n* (%)
			Total[b]	Influenza A	Influenza B	
Smoking history[i]						
Never-smoker	3805 (43.6%)	301 (49.7%)	580 (46.4%)	539 (45.8%)	43 (58.1%)	282 (50.9%)
Ex-smoker	3099 (35.5%)	198 (32.7%)	375 (30.0%)	362 (30.8%)	13 (17.6%)	188 (33.9%)
Occasional smoker	70 (0.8%)	3 (0.5%)	7 (0.6%)	6 (0.5%)	1 (1.4%)	3 (0.5%)
Daily smoker	1386 (15.9%)	63 (10.4%)	219 (17.5%)	210 (17.8%)	9 (12.2%)	50 (9.0%)
Missing	370 (4.2%)	41 (6.8%)	68 (5.4%)	60 (5.1%)	8 (10.8%)	31 (5.6%)
Long-term care facility residence, yes	942 (10.8%)	85 (14.0%)	125 (10.0%)	124 (10.5%)	1 (1.4%)	77 (13.9%)

COPD chronic obstructive pulmonary disease, *COVID-19* coronavirus disease 2019, *n or N* number, *RSV* respiratory syncytial virus, *SARI* severe acute respiratory infection, *SARS-CoV-2* severe acute respiratory syndrome coronavirus 2

[a]Test results may include co-infections; categories are not mutually exclusive

[b]Two patients had co-infection of influenza A + influenza B

[c]Subtype of RSV was not consistently tested throughout the study. Therefore, only the total number of RSV cases is provided

[d]Note that a patient could have multiple specimen collection dates for different pathogens; the earliest specimen date was considered

[e]Chronic conditions defined in Supplementary Table S3

[f]Any of the following diagnostic codes [International Classification of Diseases, 10th revision (ICD-10)]: B20-B24, D80–84, D89, Z94. Excluding disorders of the immune system, which do not lead to immunosuppression (e.g. some autoimmune conditions), adequately treated human immunodeficiency virus (HIV) infection

[g]Combination of chronic condition categories: immunodeficiency or organ transplant, and cancer

[h]COVID-19 and influenza—vaccine received ≥ 14 days before SARI symptom onset during the same epidemiological season. Pneumococcal vaccine received ≥ 14 days before symptom onset (any season)

[i]Smoking includes cigarettes, cigars, pipes, hookah, vaping, and e-cigarettes

overnight stay (n = 18), specimen collection outside the study period (n = 3), or were missing laboratory testing for SARS-CoV-2, influenza, or RSV (n = 208). The final analysis included 8730 patients (97.0% of those enrolled), with > 80.0% enrolled from Spanish sites (Table 1). Among these participants, 14.3% (n = 1249) had laboratory-confirmed influenza A, 6.9% (n = 606) SARS-CoV-2, 6.4% (n = 554) RSV, and 0.8% (n = 74) influenza B. The median age of participants was 77 years (lower–upper quartile: 65–86 years). Participants aged ≥ 75 years accounted for the majority of RSV (67.0%), SARS-CoV-2 (63.0%), and influenza A (55.0%) cases, while influenza B cases were younger (24.0% aged ≥ 75 years; 45.0% aged 18–49 years) (Table 1). Sex distribution was similar across pathogens, with females comprising 49.7%, 51.6%, and 56.0% of SARS-CoV-2, influenza, and RSV cases, respectively. Long-term care facility residence was reported for 11.0% of participants overall and was most common among SARS-CoV-2 and RSV cases (both 14.0%), consistent with the older age distribution of these groups (Table 1).

Weekly influenza and RSV test positivity rates were highest during December–January, reaching 42.3% and 19.6%, respectively (Fig. 1). By contrast, SARS-CoV-2 circulation showed a distinct summer peak, with test positivity of 27.5% in July 2025 (Fig. 1).

Symptoms

Cough (78.0%) and shortness of breath (79.0%) were the most prevalent symptoms among all hospitalised SARI patients (Fig. 2). Among RSV cases, cough (87.0%) and shortness of breath (85.0%) were most frequent, while fever was reported in only 43.0% of cases. Sudden loss of smell or taste was infrequent across all pathogen groups, including SARS-CoV-2 (3.5%). The median interval from symptom onset to specimen collection was 3 days across most pathogens, except for RSV and influenza B (4.0 and 4.5 days, respectively).

Comorbidities

Over half (52.0%) of the patients had ≥ 3 chronic conditions (Table 1). Multimorbidity was most prevalent among RSV and SARS-CoV-2 cases, with 62.0% and 55.0%, respectively, having ≥ 3 chronic conditions. Cardiovascular disease (43.0%) and hypertension (59.0%) were the most common comorbidities overall, with the highest prevalence observed among RSV (55.0% and 69.0%) and SARS-CoV-2 cases (50.0% and 64.0%). Immune-related conditions (cancer and/or immunodeficiency) were most frequent in SARS-CoV-2 positive patients (28.0%), while obesity and type 2 diabetes prevalence were highest among RSV cases (26.0% and 31.0%) (Table 1).

Treatment and Clinical Outcomes

Nearly all hospitalised SARI patients received medication during their SARI hospitalisation episode (Table 2). Among those who received medication, antibiotics were the most frequently administered medication (87.0%) across all pathogen groups, with more than three-quarters of all pathogen-specific SARI cases receiving antibiotics during hospitalisation (79.2% for SARS-CoV-2, 82.2% for influenza A/B, and 83.3% for RSV). Corticosteroids were also frequently used (68.0%), particularly among RSV cases (78.8%) and influenza A cases (70.0%). Use of antiviral medication differed by pathogen, administered to 74.8% of influenza A and 59.5% of SARS-CoV-2 cases, but was rarely used in RSV (6.4%) cases. Monoclonal antibodies or other immunomodulators were infrequently administered (< 1.0% across all pathogens).

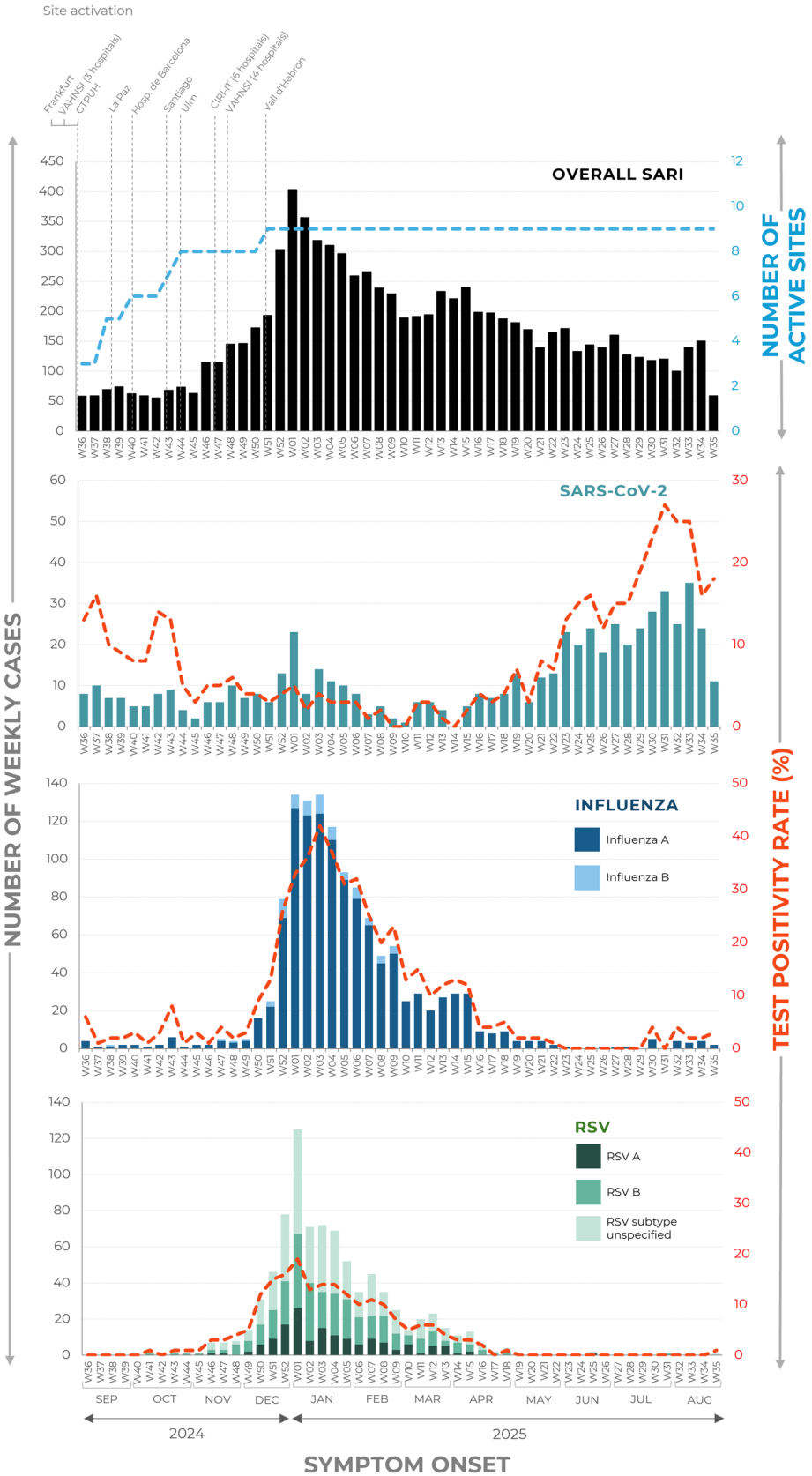
Oxygen therapy was the most common form of respiratory support, and its use varied by pathogen (Table 2). RSV cases commonly required oxygen therapy (72.0%), followed by influenza A (69.7%) and SARS-CoV-2 (58.9%). Non-invasive ventilation was most common among RSV cases (12.3%). The highest in-hospital mortality was observed among RSV cases (9.4%), followed by SARS-CoV-2 (8.2%), and the lowest for influenza A and B (4.5% and 4.1%, respectively).

Sensitivity analyses showed that sites enrolling participants directly from ICUs had higher rates of non-invasive (13.5% vs 9.1%) and invasive mechanical ventilation (1.4% vs. 0.5%) (Supplementary Table S4 and Supplementary Table S5). However, overall patterns of medication received across pathogen groups remained similar. Within sites with direct ICU enrolment, mortality among SARS-CoV-2 patients was higher than that observed among RSV patients (12.5% vs. 11.7%) (Supplementary Table S4).

Additional Pathogens of Interest

Testing for additional respiratory pathogens of interest varied substantially among participants (Table 3) and by included sites (Supplementary Table S2). Most participants were tested for adenovirus (75.6%), PIV (75.6%), HMPV (75.2%), HRV/HEV (75.3%), and RSV subtypes A and B (both 72.3%). Fewer were tested for human bocavirus (67.1%) and other human coronaviruses (30.5%) (Table 3).

Among those tested, HRV/HEV was most frequently detected (n = 908; 13.8%), followed by RSV B (n = 302; 4.8%), HMPV (n = 273; 4.2%), and PIV (n = 211; 3.2%) (Table 3). RSV A (n = 151; 2.4%) was detected in approximately half as many patients as RSV B. Adenovirus (n = 57; 0.9%) and bocavirus (n = 7; 0.1%) were rarely identified among those tested. Similar to SARS-CoV-2, influenza, and RSV, detections of these additional pathogens predominantly occurred among older adults (Table 3), with adults aged ≥


Site activation
Frankfurt
VAHNSI (3 hospitals)
GTPUH
La Paz
Hosp. de Barcelona
Santiago
Ulm
CliPi-IT (6 hospitals)
VAHNSI (4 hospitals)
Vall d'Hebron
OVERALL SARI
NUMBER OF ACTIVE SITES
SARS-CoV-2
INFLUENZA
Influenza A
Influenza B
RSV
RSV A
RSV B
RSV subtype unspecified
NUMBER OF WEEKLY CASES
TEST POSITIVITY RATE (%)
SEP
OCT
NOV
DEC
JAN
FEB
MAR
APR
MAY
JUN
JUL
AUG
2024
2025
SYMPTOM ONSET

◂**Fig. 1** Number of SARI, SARS-CoV-2, influenza (A/B), and RSV cases by week of symptom onset. id.DRIVE, 1 September 2024–31 August 2025. *CIRI-IT* Centro Interuniversitario per la Ricerca sull'Influenza e le altre Infezioni Trasmissibili, *GTPUH* German Trias I Pujol University Hospital, *RSV* respiratory syncytial virus, *SARI* severe acute respiratory infection, *SARS-CoV-2* severe acute respiratory syndrome coronavirus 2, *VAHNSI*, Valencia Hospital Surveillance Network for the Study of Influenza and Other Infectious Diseases, *W* week

75 years representing 70.9% of RSV B, 61.5% of HMPV, and 61.0% of RSV A and PIV detections.

Symptom profiles differed between pathogens. Fever was reported in 64.9% of adenovirus cases, and approximately half of HMPV (50.9%) and other human coronaviruses (48.9%) cases (Fig. 2), but was less common among PIV (42.7%) and least common among those with RSV B (41.7%). Cough and shortness of breath were frequently reported across all positive cases, with proportions exceeding 80.0% for most pathogens, including HRV/HEV, HMPV, RSV A, RSV B, and PIV.

Among cases of additional respiratory pathogens of interest, multimorbidity was common: more than half of patients with HMPV (55.7%), PIV (61.1%), HRV/HEV (51.9%), RSV A (67.0%), and RSV B (60.0%) had ≥ 3 chronic conditions (Table 3). Similar to the main pathogens, among cases of additional pathogens of interest, cardiovascular disease and hypertension were the most frequent comorbidities, with the highest prevalence observed among RSV A (58.9% and 70.2%) and HMPV (49.5% and 63.7%) cases.

Among participants tested for each respective additional pathogen, in-hospital mortality was highest among cases of RSV B (9.9%) and RSV A (8.6%), followed by other coronaviruses (8.7%), and PIV (8.0%) (Table 4). Among participants with a positive test result for SARS-CoV-2, influenza A/B, or RSV (n = 2363), only 46 (< 2.0%) had co-infection with ≥ 1 of these viruses (Supplementary Table S6).

DISCUSSION

In this multicentre hospital-based SARI surveillance study across 9 study sites covering 17 European hospitals, we described the pathogen-specific clinical and epidemiological characteristics, treatment patterns, and outcomes of adult patients hospitalised with SARI during the 2024–2025 season. By linking laboratory-confirmed respiratory pathogens with patient-level clinical and outcome data, our analysis provides a descriptive characterisation of viral SARI cases within the id.DRIVE network. These insights may complement existing European surveillance systems, which primarily report aggregated or pathogen-specific trends, although we did not formally compare the performance of our dataset with routine hospital data or other surveillance systems.

Clear seasonal patterns were observed in test positivity rates during the study period. Influenza A was the most frequently detected virus, with test positivity peaking in December–January, consistent with seasonal influenza activity reported across Europe for the same period [15]. Our observed January 2025 peak in influenza test positivity (42.3%) was similar to the pooled peak of 39.2% reported for the same countries by ECDC's hospital-based SARI virological surveillance (ERVISS) [16].

Observed trends in RSV and SARS-CoV-2 activity also aligned with reported patterns in Europe [16]. RSV exhibited a winter peak in test positivity, while SARS-CoV-2 presented a winter wave followed by a higher summer peak in July–August. This is consistent with SARS-CoV-2 patterns reported in the post-Omicron period [16, 17] and underscores the need for sustained, year-round monitoring of respiratory viruses.

Comparisons of comorbidity patterns across pathogen groups were descriptive and intended to characterise the underlying clinical profile of hospitalised patients, not to evaluate pathogen-specific risk of infection. Across pathogens, SARI hospitalisations predominantly affected older adults with a high burden of chronic comorbidities. RSV cases tended to be in the most clinically vulnerable patient group, with the highest proportion of cases aged ≥ 75 years and the

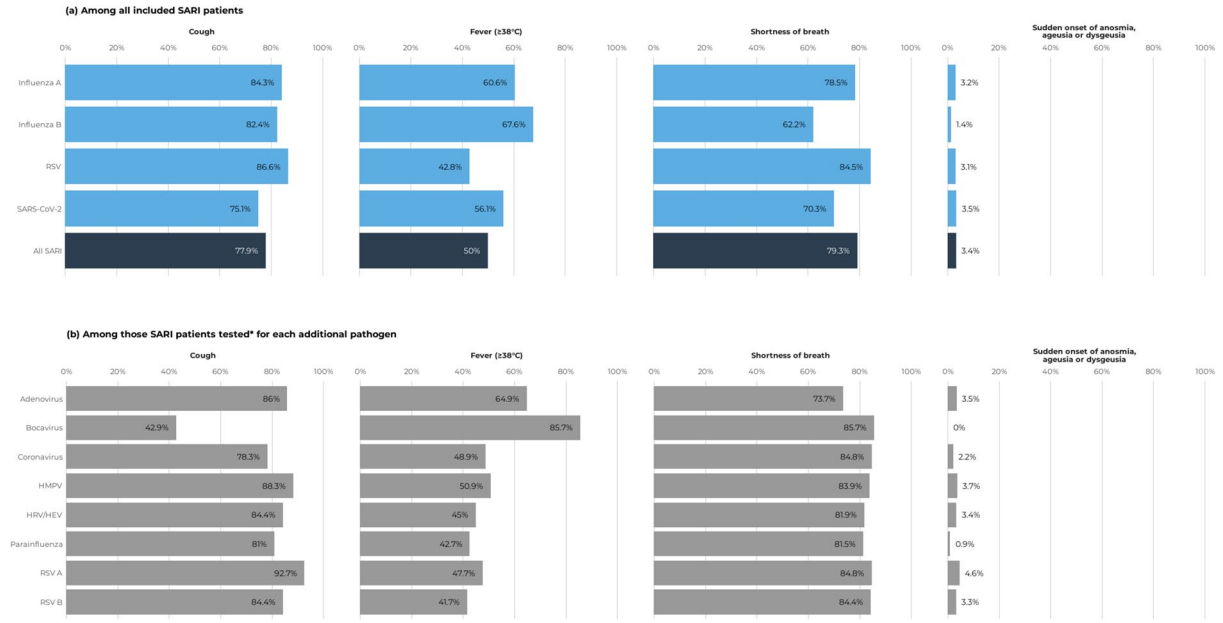


Fig. 2 Symptoms reported among SARI patients, by pathogen of interest. id.DRIVE, 1 September 2024–31 August 2025. *Not all SARI patients were tested for these pathogens; therefore, they were not included in the comparisons with all SARI patients and test-positive cases for influenza, RSV (all), and SARS-CoV-2 above. *HEV* human enterovirus, *HMPV* human metapneumovirus, *HRV* human rhinovirus, *RSV* respiratory syncytial virus, *SARI* severe acute respiratory infection, *SARS-CoV-2* severe acute respiratory syndrome coronavirus 2

highest prevalence of multimorbidity. In-hospital mortality among RSV cases exceeded that of other pathogens, consistent with previous studies demonstrating increased risk of severe RSV disease among older adults with chronic conditions [18]. The 9.4% in-hospital case-fatality rate observed is within the range reported for older adults [19, 20], although slightly higher than a recent systematic review of Europe-specific estimates [20]. This may be due to our restriction to SARI hospitalisations, which capture more severe illness than broader acute respiratory illness (ARI)-based studies, which generally report lower in-hospital case-fatality rates, as reflected in pooled estimates from recent meta-analyses [5].

Fever was least frequently reported among RSV cases relative to influenza and SARS-CoV-2, consistent with evidence that RSV often presents as afebrile or with low-grade fever in older adults [21]. This finding supports better use of ARI and modified SARI case definitions that do not require fever when estimating RSV-associated hospitalisation burden [22]. Adopting such case definitions can improve RSV surveillance sensitivity and provide more accurate estimates of RSV-associated burden, particularly when relying on surveillance data to estimate RSV disease burden [22, 23].

SARS-CoV-2-positive patients were predominantly older adults with multiple comorbidities; 80.0% had ≥ 2 chronic conditions, consistent with recent U.S. and European surveillance estimates demonstrating persistent risk of severe COVID-19 in this population [24]. Influenza A-positive patients similarly involved older adults with underlying conditions consistent with findings from other European studies [25]. Although influenza B was detected less frequently than influenza A, cases tended to occur in a younger population, underscoring how influenza B can also cause severe disease requiring hospitalisation, even among adults under 65 years [26].

The proportion of patients with confirmed viral pathogens who received antibiotics during hospitalisation for SARI was higher than reported in other European studies and exceeds

Table 2 Treatment and clinical outcomes among SARI patients, by SARS-CoV-2, influenza, and RSV test results. id. DRIVE, 1 September 2024–31 August 2025

	All SARI patients *n* (%)	SARS-CoV-2 cases *n* (%)	Influenza cases *n* (%)			RSV cases[b] *n* (%)
			Total	Influenza A	Influenza B	
Total patients[a](*N*)	8730	606	1249	1177	74	554
Any medications received during hospital stay for the management of SARI episode						
No	74 (0.8%)	6 (1.0%)	7 (0.6%)	6 (0.5%)	1 (1.4%)	3 (0.5%)
Yes	8656 (99.2%)	600 (99.0%)	1242 (99.4%)	1171 (99.5%)	73 (98.6%)	551 (99.5%)
Type of medication received during hospital stay for the management of SARI episode						
Antibiotics	7593 (87.7%)	475 (79.2%)	1021 (82.2%)	963 (82.2%)	60 (82.2%)	459 (83.3%)
Antiviral drug(s)	1389 (16.0%)	357 (59.5%)	923 (74.3%)	876 (74.8%)	49 (67.1%)	35 (6.4%)
Corticosteroid(s)	5888 (68.0%)	385 (64.2%)	859 (69.2%)	820 (70.0%)	41 (56.2%)	434 (78.8%)
Anti-SARS-CoV-2 antibodies	3 (0.0%)	3 (0.5%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Other monoclonal antibodies and immunomodulators	9 (0.1%)	1 (0.2%)	1 (0.1%)	0 (0.0%)	1 (1.4%)	2 (0.4%)
None of the above	1061 (12.3%)	37 (6.2%)	153 (12.3%)	144 (12.3%)	9 (12.3%)	77 (14.0%)
No information	2 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.2%)
Any medications received prior to hospital admission for the management of current SARI episode						
No	4839 (55.4%)	326 (53.8%)	742 (59.4%)	703 (59.7%)	40 (54.1%)	320 (57.8%)
Yes	3891 (44.6%)	280 (46.2%)	507 (40.6%)	474 (40.3%)	34 (45.9%)	234 (42.2%)
Type of medication received prior to hospital admission for the management of current SARI episode						
Antibiotics	1201 (30.9%)	57 (20.4%)	154 (30.4%)	144 (30.4%)	10 (29.4%)	83 (35.5%)
Antiviral drug(s)	28 (0.7%)	4 (1.4%)	16 (3.2%)	14 (3.0%)	2 (5.9%)	0 (0.0%)
Corticosteroid(s)	727 (18.7%)	22 (7.9%)	101 (19.9%)	96 (20.3%)	5 (14.7%)	60 (25.6%)
Anti-SARS-CoV-2 antibodies	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Other monoclonal antibodies and immunomodulators	6 (0.2%)	0 (0.0%)	2 (0.4%)	2 (0.4%)	0 (0.0%)	2 (0.9%)
None of the above	2278 (58.5%)	209 (74.6%)	293 (57.8%)	272 (57.4%)	22 (64.7%)	127 (54.3%)
SARI severity level						
Hospital admission without ICU admission	7842 (89.8%)	543 (89.6%)	1145 (91.7%)	1080 (91.8%)	67 (90.5%)	486 (87.7%)

Table 2 continued

	All SARI patients *n* (%)	SARS-CoV-2 cases *n* (%)	Influenza cases *n* (%)			RSV cases[b] *n* (%)
			Total	Influenza A	Influenza B	
ICU admission	293 (3.4%)	12 (2.0%)	47 (3.8%)	44 (3.7%)	3 (4.1%)	16 (2.9%)
In-hospital death	588 (6.8%)	50 (8.2%)	55 (4.4%)	52 (4.5%)	3 (4.1%)	52 (9.4%)
Missing	7 (0.1%)	1 (0.1%)	2 (0.2%)	1 (0.1%)	1 (1.3%)	0 (0.0%)
Respiratory support[c]						
No respiratory support	1859 (21.3%)	177 (29.2%)	262 (21.0%)	235 (20.0%)	27 (36.5%)	77 (13.9%)
Oxygen therapy (e.g. nasal cannula, mask)	5809 (66.5%)	357 (58.9%)	861 (68.9%)	820 (69.7%)	43 (58.1%)	399 (72.0%)
Non-invasive ventilation[d]	964 (11.0%)	65 (10.7%)	111 (8.9%)	109 (9.3%)	2 (2.7%)	68 (12.3%)
Invasive mechanical ventilation[e]	76 (0.9%)	4 (0.7%)	13 (1.0%)	11 (0.9%)	2 (2.7%)	8 (1.4%)
ECMO	9 (0.1%)	0 (0.0%)	1 (0.1%)	1 (0.1%)	0 (0.0%)	0 (0.0%)
Missing	13 (0.1%)	3 (0.5%)	1 (0.1%)	1 (0.1%)	0 (0.0%)	2 (0.4%)
Length of hospital stay						
Number of days, median (lower, upper quartile)	6 (4, 10)	6 (4, 10)	6 (4, 9)	6 (4, 9)	5 (3, 8.75)	7 (4, 10)
1–3 days	1669 (19.1%)	125 (20.6%)	267 (21.4%)	241 (20.5%)	26 (35.1%)	87 (15.7%)
4–6 days	2705 (31.0%)	195 (32.2%)	405 (32.4%)	383 (32.5%)	23 (31.1%)	182 (32.9%)
7–13 days	3112 (35.6%)	197 (32.5%)	420 (33.6%)	406 (34.5%)	15 (20.3%)	200 (36.1%)
14–27 days	1006 (11.5%)	70 (11.6%)	134 (10.7%)	125 (10.6%)	9 (12.2%)	68 (12.3%)
28–41 days	159 (1.8%)	10 (1.7%)	13 (1.0%)	13 (1.1%)	0 (0.0%)	11 (2.0%)
≥ 42 days	79 (0.9%)	9 (1.5%)	10 (0.8%)	9 (0.8%)	1 (1.4%)	6 (1.1%)
Discharge destination						
Home	6074 (69.6%)	408 (67.3%)	926 (74.1%)	870 (73.9%)	58 (78.4%)	369 (66.6%)
Home with nursing care	776 (8.9%)	44 (7.3%)	106 (8.5%)	100 (8.5%)	6 (8.1%)	48 (8.7%)
Transfer to another hospital	55 (0.6%)	4 (0.7%)	9 (0.7%)	7 (0.6%)	2 (2.7%)	2 (0.4%)
Rehabilitation centre	138 (1.6%)	19 (3.1%)	26 (2.1%)	24 (2.0%)	2 (2.7%)	15 (2.7%)

Table 2 continued

	All SARI patients *n* (%)	SARS-CoV-2 cases *n* (%)	Influenza cases *n* (%)			RSV cases[b] *n* (%)
			Total	Influenza A	Influenza B	
Nursing home or other long-term care facility	606 (6.9%)	46 (7.6%)	90 (7.2%)	89 (7.6%)	1 (1.4%)	53 (9.6%)
Other	394 (4.5%)	20 (3.3%)	29 (2.3%)	27 (2.3%)	2 (2.7%)	8 (1.4%)
Missing	687 (7.9%)	65 (10.7%)	63 (5.0%)	60 (5.1%)	3 (4.1%)	59 (10.6%)

ECMO extracorporeal membrane oxygenation, *ICU* Intensive care unit, *n or N* number, *RSV* respiratory syncytial virus, *SARI* severe acute respiratory infection, *SARS-CoV-2* severe acute respiratory syndrome coronavirus 2

[a]Test results may include co-infections; categories are not mutually exclusive

[b]Subtype of RSV was not consistently tested throughout the study. Therefore, only the total number of RSV cases is provided

[c]When a patient received multiple types of respiratory support, the highest level was considered

[d]Ventilatory support without tracheal intubation, e.g. high-flow nasal oxygen, continuous positive airway pressure (CPAP), or bi-level intermittent positive airway pressure

[e]Ventilatory support with tracheal intubation

Only four hospitals (responsible for 44% included SARI patients) directly recruited patients from ICU. Unless otherwise stated, continuous variables presented as medians (upper quartile, lower quartile)

the expected prevalence of both bacterial co-infection and secondary infections among hospitalised viral pathogen-specific SARI patients [27, 28]. Although antibiotics are often started empirically for suspected bacterial infection in viral SARI cases, they are frequently continued even after confirmation of a viral pathogen and negative bacterial testing. This reflects a broader tendency toward empiric antibiotic use in viral respiratory infections and reinforces the need for strengthened diagnostic stewardship and review practices to prevent avoidable antimicrobial exposure and subsequent antimicrobial resistance risks [29]. Clinician education, earlier rapid viral diagnosis, structured review of antibiotic treatment after admission, institution-specific guidelines for the management of common infections, and use of adjunctive biomarkers such as procalcitonin may help to reduce unnecessary antibiotic use [30–32].

Testing for additional respiratory viruses beyond influenza, SARS-CoV-2, and RSV was not systematic and was performed in a subset of participating hospitals. However, this testing identified a range of additional respiratory viruses, including HRV/HEV, HMPV, parainfluenza viruses, and adenoviruses, predominantly among older adults with high levels of multimorbidity, similar to patient profiles reported in studies with routine testing performed [33, 34]. HRV/HEV was the most frequently detected additional pathogen, in line with previous reports identifying HRV as a common cause of SARI in hospitalised adults [35]. Although often considered less virulent than other respiratory viruses, the observed in-hospital mortality among HRV/HEV cases supports evidence that HRV can be associated with clinically significant disease in older adults with comorbidity [36]. HMPV was detected less frequently and predominantly in older adults with multiple comorbidities. In-hospital mortality among HMPV cases was lower than that reported in similar studies of medically attended cases [37], which may partly reflect the inclusion of hospitals that did not enrol participants from the ICU, thereby missing the most severely ill patients.

Table 3 Characteristics of SARI patients by adenovirus, bocavirus, coronavirus, HMPV, human rhinovirus/enterovirus, parainfluenza, and RSV A/B. id.DRIVE, 1 September 2024–31 August 2025

	Adenovirus *n* (%)	Bocavirus *n* (%)	Coronavirus *n* (%)	HMPV *n* (%)	HRV/HEV *n* (%)	Parainfluenza *n* (%)	RSV A *n* (%)	RSV B *n* (%)
Number of patients tested	6598	5854	2663	6568	6574	6604	6319	6319
Total[a] (*N*)	57	7	92	273	908	211	151	302
Sex								
Male	21 (36.8%)	3 (42.9%)	47 (51.1%)	109 (39.9%)	434 (47.8%)	97 (46.0%)	63 (41.7%)	126 (41.7%)
Female	36 (63.2%)	4 (57.1%)	45 (48.9%)	164 (60.1%)	474 (52.2%)	114 (54.0%)	88 (58.3%)	176 (58.3%)
Age, years, median (lower, upper quartile)	70 (57, 83)	66 (63, 74)	73 (62.75, 81.25)	78 (69, 88)	76 (64, 85)	78 (66.5, 87)	80 (70.5, 87.5)	80 (71.25, 88)
Age group								
18–49 years	9 (15.8%)	0 (0.0%)	8 (8.7%)	21 (7.7%)	84 (9.3%)	12 (5.7%)	4 (2.6%)	9 (3.0%)
50–59 years	8 (14.0%)	1 (14.3%)	9 (9.8%)	19 (7.0%)	90 (9.9%)	19 (9.0%)	6 (4.0%)	19 (6.3%)
60–74 years	19 (33.3%)	4 (57.1%)	35 (38.0%)	65 (23.8%)	246 (27.1%)	51 (24.2%)	49 (32.5%)	60 (19.9%)
≥ 75 years	21 (36.8%)	2 (28.6%)	40 (43.5%)	168 (61.5%)	488 (53.7%)	129 (61.1%)	92 (60.9%)	214 (70.9%)
Country								
Germany	0 (0.0%)	0 (0.0%)	2 (2.2%)	8 (2.9%)	9 (1.0%)	4 (1.9%)	2 (1.3%)	3 (1.0%)
Italy	5 (8.8%)	2 (28.6%)	20 (21.7%)	26 (9.5%)	62 (6.8%)	22 (10.4%)	21 (13.9%)	12 (4.0%)
Spain	52 (91.2%)	5 (71.4%)	70 (76.1%)	239 (87.5%)	837 (92.2%)	185 (87.7%)	128 (84.8%)	287 (95.0%)
Symptom onset to specimen collection[b]								
Number of days, median (lower, upper quartile)	3 (2, 5)	3 (2.5, 4)	3 (2, 5)	4 (2, 6)	4 (2, 7)	3 (2, 6)	4 (2, 5)	4 (2, 6)
Number of chronic conditions								
0	6 (10.5%)	1 (14.3%)	9 (9.8%)	14 (5.1%)	58 (6.4%)	12 (5.7%)	10 (6.6%)	6 (2.0%)
1	17 (29.8%)	2 (28.6%)	12 (13.0%)	42 (15.4%)	170 (18.7%)	30 (14.2%)	16 (10.6%)	30 (9.9%)
2	5 (8.8%)	2 (28.6%)	25 (27.2%)	64 (23.4%)	206 (22.7%)	40 (19.0%)	24 (15.9%)	78 (25.8%)
≥ 3	28 (49.1%)	2 (28.6%)	45 (48.9%)	152 (55.7%)	471 (51.9%)	129 (61.1%)	101 (66.9%)	182 (60.3%)
Number of chronic conditions[c], median (lower, upper quartile)	3 (1, 4)	2 (1, 3)	2 (2, 4)	3 (2, 4)	3 (1, 4)	3 (2, 4)	3 (2, 4)	3 (2, 4)
Chronic condition[c]								
Asthma	7 (12.3%)	0 (0.0%)	9 (9.8%)	30 (11.0%)	97 (10.7%)	28 (13.3%)	14 (9.3%)	49 (16.2%)
Lung disease	15 (26.3%)	4 (57.1%)	36 (39.1%)	83 (30.4%)	321 (35.4%)	82 (38.9%)	55 (36.4%)	105 (34.8%)
COPD	8 (53.3%)	2 (50%)	25 (69.4%)	52 (62.7%)	213 (66.4%)	59 (72%)	37 (67.3%)	69 (65.7%)
Cardiovascular disease	23 (40.4%)	2 (28.6%)	37 (40.2%)	135 (49.5%)	352 (38.8%)	106 (50.2%)	89 (58.9%)	164 (54.3%)
Congestive heart failure	10 (43.5%)	0 (0.0%)	14 (37.8%)	43 (31.9%)	92 (26.1%)	37 (34.9%)	20 (22.5%)	42 (25.6%)
Hypertension	24 (42.1%)	1 (14.3%)	55 (59.8%)	174 (63.7%)	518 (57.0%)	123 (58.3%)	106 (70.2%)	207 (68.5%)
Chronic renal disease	8 (14.0%)	1 (14.3%)	21 (22.8%)	47 (17.2%)	152 (16.7%)	52 (24.6%)	32 (21.2%)	79 (26.2%)
End-stage kidney disease	3 (37.5%)	0 (0.0%)	5 (23.8%)	18 (38.3%)	34 (22.4%)	15 (28.8%)	0 (0.0%)	0 (0.0%)

Table 3 continued

	Adenovirus *n* (%)	Bocavirus *n* (%)	Coronavirus *n* (%)	HMPV *n* (%)	HRV/HEV *n* (%)	Parainfluenza *n* (%)	RSV A *n* (%)	RSV B *n* (%)
Chronic liver disease	5 (8.8%)	0 (0.0%)	8 (8.7%)	15 (5.5%)	75 (8.3%)	14 (6.6%)	10 (6.6%)	12 (4.0%)
End-stage liver disease	0 (0.0%)	NA	0 (0.0%)	1 (6.7%)	5 (6.7%)	1 (7.1%)	0 (0.0%)	0 (0.0%)
Obesity	15 (26.3%)	0 (0.0%)	8 (8.7%)	85 (31.1%)	205 (22.6%)	53 (25.1%)	40 (26.5%)	77 (25.5%)
Type 1 diabetes	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.4%)	6 (0.7%)	0 (0.0%)	0 (0.0%)	3 (1.0%)
Type 2 diabetes	16 (28.1%)	1 (14.3%)	26 (28.3%)	73 (26.7%)	275 (30.3%)	68 (32.2%)	49 (32.5%)	84 (27.8%)
Neurological disorders	15 (26.3%)	0 (0.0%)	19 (20.7%)	63 (23.1%)	187 (20.6%)	44 (20.9%)	33 (21.9%)	71 (23.5%)
Cancer	13 (22.8%)	2 (28.6%)	26 (28.3%)	39 (14.3%)	176 (19.4%)	50 (23.7%)	28 (18.5%)	55 (18.2%)
Solid tumour	11 (84.6%)	1 (50.0%)	12 (46.2%)	23 (59.0%)	115 (65.3%)	34 (68.0%)	16 (57.1%)	32 (58.2%)
Haematological cancer	3 (23.1%)	1 (50.0%)	14 (53.8%)	16 (41.0%)	61 (34.7%)	18 (36.0%)	11 (39.3%)	23 (41.8%)
Immunodeficiency[d]	6 (10.5%)	1 (14.3%)	16 (17.4%)	17 (6.2%)	84 (9.3%)	26 (12.3%)	7 (4.6%)	20 (6.6%)
Cancer and/or immunodeficiency[e]	15 (26.3%)	2 (28.6%)	33 (35.9%)	50 (18.3%)	222 (24.4%)	59 (28.0%)	30 (19.9%)	64 (21.2%)
Vaccination status[f] (yes)								
COVID-19	36 (63.2%)	5 (71.4%)	49 (53.3%)	178 (65.2%)	522 (57.5%)	125 (59.2%)	94 (62.3%)	211 (69.9%)
Influenza	29 (50.9%)	4 (57.1%)	41 (44.6%)	170 (62.3%)	461 (50.8%)	107 (50.7%)	101 (66.9%)	173 (57.3%)
Pneumococcal disease	21 (36.8%)	5 (71.4%)	48 (52.2%)	142 (52.0%)	419 (46.1%)	102 (48.3%)	78 (51.7%)	167 (55.3%)
Smoking history[g]								
Never-smoker	24 (42.1%)	3 (42.9%)	34 (37.0%)	155 (56.8%)	412 (45.4%)	99 (46.9%)	71 (47.0%)	167 (55.3%)
Ex-smoker	20 (35.1%)	3 (42.9%)	31 (33.7%)	90 (33.0%)	295 (32.5%)	75 (35.5%)	57 (37.7%)	91 (30.1%)
Occasional smoker	0 (0.0%)	0 (0.0%)	0 (0.0%)	5 (1.8%)	6 (0.7%)	3 (1.4%)	2 (1.3%)	1 (0.3%)
Daily smoker	9 (15.8%)	1 (14.3%)	18 (19.6%)	12 (4.4%)	179 (19.7%)	26 (12.3%)	13 (8.6%)	24 (7.9%)
Missing	4 (7.0%)	0 (0.0%)	9 (9.8%)	11 (4.0%)	16 (1.8%)	8 (3.8%)	8 (5.3%)	19 (6.3%)
Long-term care facility residence, yes	6 (10.5%)	1 (14.3%)	11 (12.0%)	33 (12.1%)	107 (11.8%)	28 (13.3%)	14 (9.3%)	48 (15.9%)

COPD chronic obstructive pulmonary disease, *COVID-19* coronavirus disease 2019, *HEV* human enterovirus, *HMPV* human metapneumovirus, *HRV* human rhinovirus, *n or N* number, *RSV* respiratory syncytial virus, *SARI* severe acute respiratory infection

[a]Test results may include co-infections; categories are not mutually exclusive

[b]Note that a patient could have multiple specimen collection dates for different pathogens; the earliest specimen date was considered

[c]Chronic conditions defined in Supplementary Table S3

[d]Any of the following diagnostic codes [International Classification of Diseases, 10th revision (ICD-10)]: B20-B24, D80–84, D89, Z94. Excluding disorders of the immune system, which do not lead to immunosuppression (e.g. some autoimmune conditions), adequately treated human immunodeficiency virus (HIV) infection

[e]Combination of chronic condition categories: immunodeficiency or organ transplant, and cancer

[f]COVID-19 and influenza vaccine received ≥ 14 days before SARI symptom onset during the same epidemiological season. Pneumococcal vaccine received ≥ 14 days before symptom onset (any season)

[g]Smoking includes cigarettes, cigars, pipes, hookah, vaping, and e-cigarettes

Table 4 Treatment and clinical outcomes among SARI patients by adenovirus, bocavirus, coronavirus, HMPV, human rhinovirus/enterovirus, parainfluenza, and RSV A/B. id.DRIVE, 1 September 2024–31 August 2025

	Adenovirus *n* (%)	**Bocavirus *n* (%)**	**Coronavirus *n* (%)**	**HMPV *n* (%)**	**HRV/HEV *n* (%)**	**Parainfluenza *n* (%)**	**RSV A *n* (%)**	**RSV B *n* (%)**
Number of patients tested	6598	5854	2663	6568	6574	6604	6319	6319
Total patients[a](*N*)	57	7	92	273	908	211	151	302
Any medications received during hospital stay for the management of SARI episode								
No	1 (1.8%)	0 (0.0%)	0 (0.0%)	1 (0.4%)	5 (0.6%)	0 (0.0%)	1 (0.7%)	2 (0.7%)
Yes	56 (98.2%)	7 (100.0%)	92 (100.0%)	272 (99.6%)	903 (99.4%)	211 (100.0%)	150 (99.3%)	300 (99.3%)
Type of medication received during hospital stay for the management of SARI episode								
Antibiotics	53 (94.6%)	7 (100.0%)	74 (80.4%)	254 (93.4%)	830 (91.9%)	193 (91.5%)	121 (80.7%)	245 (81.7%)
Antiviral drug(s)	3 (5.4%)	0 (0.0%)	6 (6.5%)	5 (1.8%)	31 (3.4%)	11 (5.2%)	9 (6.0%)	14 (4.7%)
Corticosteroid(s)	40 (71.4%)	4 (57.1%)	57 (62.0%)	201 (73.9%)	696 (77.1%)	150 (71.1%)	119 (79.3%)	237 (79.0%)
Anti-SARS-CoV-2 antibodies	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Other monoclonal antibodies and immunomodulators	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.1%)	0 (0.0%)	1 (0.7%)	0 (0.0%)
None of the above	3 (5.4%)	1 (14.3%)	9 (9.8%)	13 (4.8%)	109 (12.1%)	16 (7.6%)	27 (18.0%)	41 (13.7%)
Any medications received prior to hospital admission for the management of current SARI episode								
No	22 (38.6%)	4 (57.1%)	31 (33.7%)	124 (45.4%)	538 (59.3%)	114 (54.0%)	83 (55.0%)	199 (65.9%)
Yes	35 (61.4%)	3 (42.9%)	61 (66.3%)	149 (54.6%)	370 (40.7%)	97 (46.0%)	68 (45.0%)	103 (34.1%)
Type of medication received prior to hospital admission for the management of current SARI episode								
Antibiotics	15 (42.9%)	2 (66.7%)	14 (23.0%)	67 (45.0%)	128 (34.6%)	24 (24.7%)	24 (35.3%)	38 (36.9%)
Antiviral drug(s)	0 (0.0%)	0 (0.0%)	1 (1.6%)	0 (0.0%)	3 (0.8%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Corticosteroid(s)	5 (14.3%)	1 (33.3%)	11 (18.0%)	34 (22.8%)	92 (24.9%)	16 (16.5%)	21 (30.9%)	26 (25.2%)

Table 4 continued

	Adenovirus *n* (%)	Bocavirus *n* (%)	Coronavirus *n* (%)	HMPV *n* (%)	HRV/HEV *n* (%)	Parainfluenza *n* (%)	RSV A *n* (%)	RSV B *n* (%)
Anti-SARS-CoV-2 antibodies	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Other monoclonal antibodies and immunomodulators	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.3%)	0 (0.0%)	2 (2.9%)	0 (0.0%)
None of the above	18 (51.4%)	1 (33.3%)	41 (67.2%)	70 (47.0%)	180 (48.6%)	64 (66.0%)	37 (54.4%)	53 (51.5%)
SARI severity level								
Hospital admission without ICU admission	51 (89.5%)	6 (85.7%)	79 (85.9%)	252 (92.3%)	826 (91.0%)	187 (88.6%)	135 (89.4%)	264 (87.4%)
ICU admission	4 (7.0%)	1 (14.3%)	5 (5.4%)	11 (4.0%)	39 (4.3%)	7 (3.3%)	3 (2.0%)	8 (2.6%)
In-hospital death	2 (3.6%)	0 (0.0%)	8 (8.7%)	10 (3.6%)	43 (4.7%)	17 (8.0%)	13 (8.6%)	30 (9.9%)
Respiratory support[b]								
No respiratory support	7 (12.3%)	1 (14.3%)	30 (32.6%)	38 (13.9%)	167 (18.4%)	40 (19.0%)	25 (16.6%)	46 (15.2%)
Oxygen therapy (e.g. nasal cannula, mask)	44 (77.2%)	5 (71.4%)	54 (58.7%)	201 (73.6%)	638 (70.3%)	154 (73.0%)	107 (70.9%)	219 (72.5%)
Non-invasive ventilation[c]	4 (7.0%)	1 (14.3%)	7 (7.6%)	30 (11.0%)	99 (10.9%)	15 (7.1%)	15 (9.9%)	33 (10.9%)
Invasive mechanical ventilation[d]	1 (1.8%)	0 (0.0%)	1 (1.1%)	3 (1.1%)	2 (0.2%)	1 (0.5%)	4 (2.6%)	3 (1.0%)
ECMO	1 (1.8%)	0 (0.0%)	0 (0.0%)	1 (0.4%)	1 (0.1%)	1 (0.5%)	0 (0.0%)	0 (0.0%)
Length of hospital stay								
Number of days, median (lower, upper quartile)	7 (5, 10)	6 (3, 11.5)	8 (5, 10)	8 (4, 12)	7 (4, 10)	7 (4, 11)	7 (5, 11)	6 (4, 10)
1–3 days	6 (10.5%)	2 (28.6%)	9 (9.8%)	49 (17.9%)	166 (18.3%)	37 (17.5%)	22 (14.6%)	55 (18.2%)
4–6 days	17 (29.8%)	2 (28.6%)	24 (26.1%)	63 (23.1%)	285 (31.4%)	58 (27.5%)	44 (29.1%)	101 (33.4%)

Table 4 continued

	Adenovirus *n* (%)	Bocavirus *n* (%)	Coronavirus *n* (%)	HMPV *n* (%)	HRV/HEV *n* (%)	Parainfluenza *n* (%)	RSV A *n* (%)	RSV B *n* (%)
7–13 days	26 (45.6%)	2 (28.6%)	40 (43.5%)	110 (40.3%)	328 (36.1%)	85 (40.3%)	59 (39.1%)	110 (36.4%)
14–27 days	7 (12.3%)	1 (14.3%)	16 (17.4%)	47 (17.2%)	106 (11.7%)	28 (13.3%)	23 (15.2%)	28 (9.3%)
≥ 28 days	1 (1.8%)	0 (0.0%)	3 (3.3%)	4 (1.5%)	23 (2.5%)	3 (1.4%)	3 (1.4%)	8 (2.6%)
Discharge destination								
Home	45 (78.9%)	4 (57.1%)	65 (70.7%)	197 (72.2%)	669 (73.7%)	156 (73.9%)	108 (71.5%)	193 (63.9%)
Home with nursing care	4 (7.0%)	1 (14.3%)	9 (9.8%)	23 (8.4%)	81 (8.9%)	9 (4.3%)	11 (7.3%)	24 (7.9%)
Transfer to another hospital	0 (0.0%)	0 (0.0%)	0 (0.0%)	2 (0.7%)	2 (0.2%)	1 (0.5%)	0 (0.0%)	2 (0.7%)
Rehabilitation centre	0 (0.0%)	0 (0.0%)	2 (2.2%)	3 (1.1%)	9 (1.0%)	3 (1.4%)	6 (4.0%)	6 (2.0%)
Nursing home or other long-term care facility	3 (5.3%)	2 (28.6%)	7 (7.6%)	25 (9.2%)	63 (6.9%)	17 (8.1%)	8 (5.3%)	39 (12.9%)
Other	3 (5.3%)	0 (0.0%)	0 (0.0%)	10 (3.7%)	36 (4.0%)	6 (2.8%)	1 (0.7%)	5 (1.7%)
Missing	2 (3.5%)	0 (0.0%)	9 (9.8%)	13 (4.8%)	48 (5.3%)	19 (9.0%)	17 (11.3%)	33 (10.9%)

ECMO extracorporeal membrane oxygenation, *HEV* human enterovirus, *HMPV* human metapneumovirus, *HRV* human rhinovirus, *ICU* intensive care unit, *n or N* number, *RSV* respiratory syncytial virus, *SARI* severe acute respiratory infection, *SARS-CoV-2* severe acute respiratory syndrome coronavirus 2

[a]Test results may include co-infections; categories are not mutually exclusive

[b]When a patient received multiple types of respiratory support, the highest level was considered

[c]Ventilatory support without tracheal intubation, e.g. high-flow nasal oxygen, continuous positive airway pressure (CPAP), or bi-level intermittent positive airway pressure

[d]Ventilatory support with tracheal intubation

Limitations

This study has several limitations. First, recruitment was staggered across participating sites, with 12 hospitals (71.0%) initiating enrolment after the start of the study period. Although all hospitals enrolled participants by 19 December 2024, this may have resulted in underrepresentation of early-season (September–November 2024) SARI cases, influencing estimates of pathogen-specific positivity, patient characteristics, and outcomes. Second, 13 (76.0%) hospitals were unable to enrol participants directly from the ICUs. While some participants could be enrolled retrospectively following ICU discharge, those who died in the ICU were likely missed, potentially leading to underestimation of disease severity and in-hospital mortality. The magnitude of this potential bias could not be quantified, and true mortality for non-ICU-enrolling sites remains uncertain. Third, the geographic distribution of enrolled participants was

heterogeneous, with most participants recruited from Spain (89.2%). Thus, the findings may have limited generalisability to other participating countries or to Europe overall.

Fourth, testing for respiratory viruses beyond influenza A and B, SARS-CoV-2, and RSV was not systematic. Although 14 (82.0%) hospitals conducted PCR testing for additional respiratory pathogens of interest, only 4 (24.0%) performed such testing for the entire study period (Supplementary Table S2). This likely led to an underestimation of positivity rates for other respiratory viruses. In sites without routine testing for additional pathogens, testing was likely performed preferentially in patients with more severe clinical presentations, potentially overrepresenting severe cases among those with confirmed infections and limiting comparability of pathogen-specific test positivity. Because the indication for testing was not systematically recorded at these sites, the magnitude of this potential selection bias could not be quantified. Also, additional co-infections were likely under-ascertained.

Fifth, RSV detection relied on collecting a single respiratory specimen. Given the lower sensitivity of RSV testing from a single sample than multispecimen approaches [7], especially in adults or later in the course of illness, some RSV infections may have been missed. Similarly, reliance on nasopharyngeal swabs alone, which were used in 51.6–51.8% of SARS-CoV-2, RSV, and influenza cases, may have reduced detection of HMPV, adenovirus, and PIVs, leading to under-ascertainment of these infections and underestimation of their relative contribution to SARI hospitalisations.

Sixth, co-infections could be reliably assessed only for influenza, SARS-CoV-2, and RSV, meaning the number of participants with documented co-infections is likely substantially underestimated. However, co-infection rates of SARS-CoV-2, influenza, and RSV are similar to those reported in more recent European studies focused on these pathogens [38], although the estimates are highly heterogeneous, making comparisons difficult [39]. Seventh, because our pathogen-specific categories included co-infections and were therefore not mutually exclusive, we were unable to perform valid direct statistical comparisons of treatment patterns or clinical outcomes between viral groups. Finally, this analysis reflects one surveillance year, limiting generalisability to other years given the variability in respiratory virus circulation by season.

Strengths

This study provides pathogen-specific epidemiological and clinical profiles of SARI patients across multiple European hospitals. Unlike existing national or European surveillance systems, which primarily report aggregate or pathogen-specific trends [12], our findings allow direct, standardised comparisons of several viral respiratory pathogens within the same clinical framework.

Furthermore, our SARI case definition did not require fever, allowing the inclusion of participants with atypical presentations. This is particularly relevant for RSV, which frequently presents without fever in older adults, thereby improving the representativeness of RSV-associated SARI across age groups and enhancing comparability between pathogens [21–23].

CONCLUSION

This multicentre SARI surveillance study across 17 European hospitals demonstrates the ongoing burden of viral respiratory pathogens among hospitalised adults, with distinct seasonal patterns and a high burden among older adults with multimorbidity. While influenza A was the most frequently detected virus, RSV showed greater healthcare utilisation and in-hospital mortality, underscoring the significant impact of RSV in older adults. Our detection of substantial numbers of SARI cases with additional viral pathogens other than SARS-CoV-2, RSV, and influenza highlights the broad viral diversity underlying SARI in hospitalised adults.

These findings emphasise the value of integrated, pathogen-specific SARI surveillance in capturing both epidemiological trends and clinical outcomes across viral respiratory

pathogens. Future work comparing our findings with other routine or surveillance sources would help define its additional value for public health monitoring and prevention planning. Strengthened sentinel networks, like id.DRIVE, are essential for guiding preventive strategies, including vaccine and treatment development, for Europe's ageing population.

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Data Availability. De-identified data supporting the results presented in this article (text, tables, figures, and appendices) are available upon request (info@iddrive.eu) in accordance with the id.DRIVE data access policy, as outlined in the id.DRIVE Governance Charter.

Declarations

Conflict of Interest. Katherine Newell, Nikki Vroom, Kok-Yew Ngew, Juan V. Hernandez-Villena, Hollie Dunford, Somsuvro Basu, and Kaatje Bollaerts are employed by P95. Kaatje Bollaerts has received consulting fees from Pfizer for conducting this study, may own P95 stocks, and reported consulting fees from AstraZeneca, CEPI, GSK, Pfizer, Valneva, and WHO. She also reported book royalties. Cátia Marques, Farid L. Khan, Hannah R. Volkman, Negar Aliabadi, Elizabeth Begier, and Jennifer L. Nguyen are employed by Pfizer and may own Pfizer stocks or stock options. Ekaterina Maslova and Chengbin Wang are employed by AstraZeneca and may own AstraZeneca stocks or stock options. Susana Otero-Romero declares consulting fees from Moderna and Sanofi, honoraria from AstraZeneca and GSK, and support from Pfizer for meeting attendance and travel. Irma Casas reports support from GSK and Pfizer for meeting attendance and travel. Elisa Martró reports grants/contracts and support for meeting attendance and travel from Gilead Sciences. Giancarlo Icardi, coordinator of the CIRI-IT network, received funding from P95 via id.DRIVE for the study. Federico Martinón-Torres reports grants/contracts paid to his institution from GSK, Pfizer, and Sanofi-AstraZeneca; consulting or advisory fees from GSK, Pfizer, Sanofi Pasteur, Janssen Pharmaceuticals, MSD, Seqirus, Biofabri, and AstraZeneca; and support for meeting attendance and travel from Pfizer, MSD, GSK, and Sanofi. He is also a member of the European Technical Advisory Group of Experts on Immunization, coordinator of the Spanish Pediatric Critical Trials Network, and coordinator of the WHO collaborating Center for Vaccine Safety of Santiago de Compostela. He is a principal investigator in randomised controlled trials of Ablynx, Abbott, Seqirus, Sanofi Pasteur MSD, Sanofi Pasteur, Cubist, Wyeth, Merck, Pfizer, Roche, Regeneron, Jansen, MedImmune, Novavax, Novartis, and GSK, with honoraria paid to his institution. Ainara Mira-Iglesias, co-PI from VAHNSI (FISABIO), reports that her foundation received funding from P95 via id.DRIVE for the study. Her institution also received a grant from the Foundation for Influenza Epidemiology via the Global Influenza Hospital Surveillance Network (GIHSN) for VAHNSI surveillance. Alejandro Orrico-Sánchez, co-PI from VAHNSI (FISABIO), reports that his foundation received funding from P95 through id.DRIVE for the study. Antoni Torres reports consulting fees and honoraria from Jansen OM Pharma, MSD, Pfizer, and Polyphor. The other authors (Andrés Antón, Beate Grüner, Carlos M. Oñoro-López, Oleguer Parés-Badell, and Wolfgang Gleiber) report no conflicts of interest.

Ethical Approval. This study was performed in accordance with the 1964 Declaration of Helsinki and its later amendments. All data were pseudonymised prior to analysis and processed in accordance with the European Union General Data Protection Regulation [GDPR; Regulation (EU) 2016/679]. The id.DRIVE surveillance protocol (EUPAS1000000012) was approved by independent ethics committees (IECs) at all participating hospitals or hospital networks: Comitato Etico Regionale della Liguria (Genoa); Comité Ético de Investigación con Medicamentos del Departamento de Salud Arnau de Vilanova (Valencia); Comité Ético de Investigación con Medicamentos del Hospital Universitari Vall d'Hebron (Barcelona); Comité de Ética de la Investigación con Medicamentos del Hospital Universitario La Paz; Ethikkommission des Fachbereichs Medizin der Goethe-Universität (Frankfurt); Ethikkommission Ulm. Individuals (or their legally acceptable representative) provided informed consent when applicable. Waiver of informed consent was authorised by local IECs for retrospective data collection for Hospital Germans Trias i Pujol, Hospital Clinic Barcelona, and Vall d'Hebron, as the tests carried out and information collected were part of routine clinical practice and epidemiological surveillance.

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REFERENCES

1. GBD 2023 Lower Respiratory Infections and Antimicrobial Resistance Collaborators. Global burden of lower respiratory infections and aetiologies, 1990-2023: a systematic analysis for the Global Burden of Disease Study 2023. Lancet Infect Dis. 2026;26(4):343-61.

2. GBD 2017 Influenza Collaborators. Mortality, morbidity, and hospitalisations due to influenza lower respiratory tract infections, 2017: an analysis for the Global Burden of Disease Study 2017. Lancet Respir Med. 2019;7(1):69–89.

3. World Health Organization. Coronavirus (COVID-19) Dashboard. 2025 Available from: https://covid19.who.int/.

4. Tin Tin Htar M, Yerramalla MS, Moïsi JC, Swerdlow DL. The burden of respiratory syncytial virus in adults: a systematic review and meta-analysis. Epidemiol Infect. 2020;148:e48.

5. Li Y, Kulkarni D, Begier E, Wahi-Singh P, Wahi-Singh B, Gessner B, et al. Adjusting for case underascertainment in estimating rsv hospitalisation burden of older adults in high-income countries: a systematic review and modelling study. Infect Dis Ther. 2023;12(4):1137–49.

6. Rozenbaum MH, Begier E, Kurosky SK, Whelan J, Bem D, Pouwels KB, et al. Incidence of respiratory syncytial virus infection in older adults: limitations of current data. Infect Dis Ther. 2023;12(6):1487–504.

7. Begier E, Aliabadi N, Ramirez JA, McGeer A, Liu Q, Carrico R, et al. Detection by Nasopharyngeal Swabs alone underestimates respiratory syncytial virus-related hospitalization incidence in adults: the multispecimen study's final analysis. J Infect Dis. 2025;232(1):e126–36.

8. Onwuchekwa C, Moreo LM, Menon S, Machado B, Curcio D, Kalina W, et al. Underascertainment of respiratory syncytial virus infection in adults due to diagnostic testing limitations: a systematic literature review and meta-analysis. J Infect Dis. 2023;228(2):173–84.

9. Meslé MMI, Sinnathamby M, Mook P, Pebody R, Lakhani A, Zambon M, et al. Seasonal and interseasonal RSV activity in the European Region during the COVID-19 pandemic from autumn 2020 to summer 2022. Influenza Other Respir Viruses. 2023;17(11):e13219.

10. Centers for Disease Control and Prevention. Influenza activity in the united states during the 2024–25 season and composition of the 2025–26 influenza vaccine. 2025 Available from: https://www.cdc.gov/flu/whats-new/2025-2026-influenza-activity.html.

11. World Health Organization. The impact of suspensions and reductions in health official development assistance on health systems - Rapid WHO country office stock take: Summary of results from 108 WHO country offices, 7 March–2 April 2025. 2025 Available from: https://cdn.who.int/media/docs/default-source/integrated-health-services-(ihs)/impact-of-suspensions-and-reductions-in-health-oda-on-health-systems.pdf?sfvrsn=7b1cafcb_7&download=true.

12. Marques DF, Kovacs D, Sanchez-Ruiz MA, Rodrigues AP, Machado A, Mazagatos C, et al. Evaluation of European severe acute respiratory infection (SARI) surveillance, 27 European countries, 2022/23. Euro Surveill. 2025;30(20):2400655.

13. European Medicines Agency. id.DRIVE surveillance study of respiratory pathogens in adults hospitalised for severe acute respiratory infection (SARI) across Europe. 2024 Available from: https://catalogues.ema.europa.eu/node/3915/administrative-details.

14. European Center for Disease Prevention and Control. EU case definitions. 2025 Available from: https://www.ecdc.europa.eu/en/all-topics/eu-case-definitions.

15. European Center for Disease Prevention and Control. Weekly communicable disease threats report, week 14, 29 March–4 April 2025. 2025 Available from: https://gpk.de/downloadp/communicable-disease-threats-report-week-14-2025.pdf.

16. European Center for Disease Prevention and Control. European respiratory virus surveillance summary. 2025 Available from: https://erviss.org/.

17. European Center for Disease Prevention and Control. Overview of respiratory virus epidemiology in

the EU/EEA, week 42, 2025. 2025 Available from: https://erviss.org/.

18. Branche AR, Falsey AR, Saiman L, Alba L, Peterson D, Wang L, et al. Respiratory syncytial virus risk assessment in adults: implications for vaccine recommendations beyond age. Open Forum Infect Dis. 2025;12(7):0faf394.

19. Nguyen-Van-Tam JS, O'Leary M, Martin ET, Heijnen E, Callendret B, Fleischhackl R, et al. Burden of respiratory syncytial virus infection in older and high-risk adults: a systematic review and meta-analysis of the evidence from developed countries. Eur Respir Rev. 2022;31(166):220105.

20. Zhang T, Reeves RM, Ma S, Miao Y, Sun S, Orrico-Sánchez A, et al. Estimating the respiratory syncytial virus-associated hospitalisation burden in older adults in European countries: a systematic analysis. BMC Med. 2025;23(1):453.

21. Korsten K, Adriaenssens N, Coenen S, Butler CC, Verheij TJM, Bont LJ, et al. World Health Organization influenza-like illness underestimates the burden of respiratory syncytial virus infection in community-dwelling older adults. J Infect Dis. 2022;226(Suppl 1):S71–8.

22. World Health Organization. Global influenza programme: RSV surveillance case definitions. 2025 Available from: https://www.who.int/teams/global-influenza-programme/global-respiratory-syncytial-virus-surveillance/case-definitions.

23. Hirve S, Crawford N, Palekar R, Zhang W. Clinical characteristics, predictors, and performance of case definition-Interim results from the WHO global respiratory syncytial virus surveillance pilot. Influenza Other Respir Viruses. 2020;14(6):647–57.

24. Taylor CA, Patel K, Pham H, Kirley PD, Kawasaki B, Meek J, et al. COVID-19-associated hospitalizations among U.S. adults aged ≥ 18 years—COVID-NET, 12 states, October 2023–April 2024. MMWR Morb Mortal Wkly Rep. 2024;73(39):869–75.

25. Rose AMC, Nicolay N, Mazagatos C, Martínez-Baz I, Launay O, De Mot L, et al. Vaccine effectiveness against influenza A in older adults and the effect of chronic conditions: results from the I-MOVE and VEBIS multicentre European hospital case–control studies, 2015/16–2023/24. BMC Med. 2025;23(1):602.

26. Naquin A, O'Halloran A, Ujamaa D, Sundaresan D, Masalovich S, Cummings CN, et al. Laboratory-confirmed influenza-associated hospitalizations among children and adults—influenza hospitalization surveillance network, United States, 2010–2023. MMWR Surveill Summ. 2024;73(6):1–18.

27. Hedberg P, Serwin K, Francesca Greco M, Pereira JPV, Juozapaite D, De Benedittis S, et al. Bacterial and viral coinfections in adult patients hospitalized with COVID-19 throughout the pandemic: a multinational cohort study in the EuCARE project. J Infect Dis. 2025;231(6):e1091–101.

28. Abelenda-Alonso G, Rombauts A, Gudiol C, Meije Y, Ortega L, Clemente M, et al. Influenza and bacterial coinfection in adults with community-acquired pneumonia admitted to conventional wards: risk factors, clinical features, and outcomes. Open Forum Infect Dis. 2020;7(3):ofaa066.

29. European Center for Disease Prevention and Control. Point prevalence survey of healthcare-associated infections and antimicrobial use in European acute care hospitals—2022–2023. 2024 Available from: https://www.ecdc.europa.eu/en/publications-data/PPS-HAI-AMR-acute-care-europe-2022-2023.

30. World Health Organization. Antimicrobial stewardship interventions: a practical guide. 2021 Available from: https://iris.who.int/server/api/core/bitstreams/86435a2d-33ac-4813-a1c6-d018ce39e439/content.

31. Lee CC, Chang JC, Mao XW, Hsu WT, Chen SY, Chen YC, et al. Combining procalcitonin and rapid multiplex respiratory virus testing for antibiotic stewardship in older adult patients with severe acute respiratory infection. J Am Med Dir Assoc. 2020;21(1):62–7.

32. Schuetz P, Wirz Y, Sager R, Christ-Crain M, Stolz D, Tamm M, et al. Effect of procalcitonin-guided antibiotic treatment on mortality in acute respiratory infections: a patient level meta-analysis. Lancet Infect Dis. 2018;18(1):95–107.

33. Philippot Q, Rammaert B, Dauriat G, Daubin C, Schlemmer F, Costantini A, et al. Human metapneumovirus infection is associated with a substantial morbidity and mortality burden in adult inpatients. Heliyon. 2024;10(13):e33231.

34. Kwok WC, Leung IS, Ho JC, Lam DC, Ip MS, Ngai SM, et al. Medical comorbidities as the independent risk factors of severe adenovirus respiratory tract infection in adults. Microorganisms. 2025;13(7):1670.

35. Minosse C, Selleri M, Zaniratti MS, Cappiello G, Longo R, Schifano E, et al. Frequency of detection of respiratory viruses in the lower respiratory tract of hospitalized adults. J Clin Virol. 2008;42(2):215–20.

36. Jain S, Self WH, Wunderink RG, Fakhran S, Balk R, Bramley AM, et al. Community-acquired

pneumonia requiring hospitalization among U.S. adults. N Engl J Med. 2015;373(5):415–27.

37. Sobanjo-Ter Meulen A, Gutierrez AV, Ruiz O, Eeuwijk J, Vroling H, Kanesa-Thasan N. The burden of human metapneumovirus (hMPV) disease in older and high-risk adults in developed countries: a systematic literature review. Infect Dis Ther. 2025;14(8):1917–33.

38. Wiechert L, Fischer C, Jörres RA, Engelhardt S, Alter P, Kahnert K, et al. Characteristics and outcomes of patients hospitalized for infection with influenza A, SARS-CoV-2 or respiratory syncytial virus in the season 2023/2024 in a large German primary care centre. Eur J Med Res. 2024;29(1):509.

39. Golpour M, Jalali H, Alizadeh-Navaei R, Talarposhti MR, Mousavi T, Ghara AAN. Co-infection of SARS-CoV-2 and influenza A/B among patients with COVID-19: a systematic review and meta-analysis. BMC Infect Dis. 2025;25(1):145.

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